

SIMATSENGINEERING

ASSESSMENT TOOL 3

NAME: M. Jayaprakash
COURSE NAME: CSA1607

REG.NO: 192425268
COURSE CODE: Data Warehousing and Data Mining

COURSE OUTCOME COVERED

CO4: Examine the applicability of clustering algorithms in real-world scenarios, presenting findings through clear reports with effective visualizations.
Assessment Tool Weightage: 20%

QUESTIONS: 05

Total Marks: 50

Annexure A

COMPARITIVE ANALYSIS

Criteria	Scope of Items (2)	Comparison Depth (2.5)	Use of Evidence (2)	Critical Thinking (2)	Clarity of Presentation (1)	Total (10)
Weightage	20%	25%	20%	20%	10%	100%
Q1						
Q2						
Q3						
Q4						
Q5						

Code of Conduct:

*M. Jayaprakash (192425268) (Name/RegNo) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.*

Signature of the student:
jayaprakash

RUBRICS FOR CALCULATION:

		Comparative analysis			
Criteria	Weightage (%)	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Scope of Items	20%	Select highly relevant items with clear scope and justification	Selects appropriate items with minor gaps	Selection is limited or partially relevant	Items are unclear or not relevant
Comparison Depth	25%	Provides detailed and comprehensive comparison across multiple aspects	Provides adequate comparison with some depth	Limited comparison with few aspects	Very minimal or no comparison
Use of Evidence	20%	Supports comparison with accurate data, examples, or references	Uses some supporting evidence with minor gaps	Limited or weak supporting evidence	No supporting evidence provided
Critical Thinking	20%	Demonstrates strong critical thinking with clear interpretation and reasoning	Shows reasonable interpretation with minor gaps	Basic interpretation; lacks depth	No meaningful interpretation
Clarity of Presentation	15%	Well-organized, clear, and logically structured presentation	Generally clear with minor issues	Somewhat unclear or poorly structured	Disorganized and difficult to understand

AnnexureA

Q1.Dataset:AssessmentTool3:ComparativeAnalysis(25%→5 Marks)

Time duration:45Minutes

AssessmentTool3 (Low)

Q1.Dataset: CustomerData

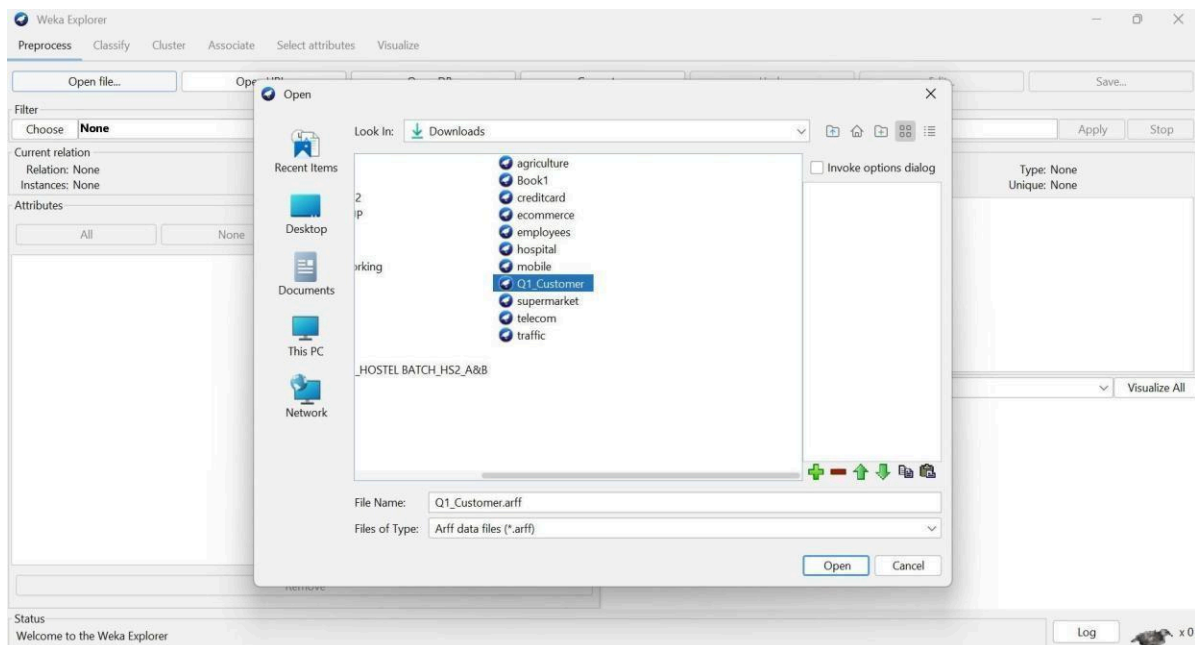
CustomerIncome(₹)SpendingScore

C1	30000	60
C2	60000	40
C3	50000	70
C4	28000	65
C5	65000	35

Question:

CompareK-meansandhierarchicalclusteringusingthisdatasetandexplain differences in performance and scalability. (BL3 – 1 Mark)

OUTPUT:



Weka Explorer

Preprocess Classify Cluster Associate Select attributes Visualize

Cluster: **HierarchicalClusterer - N 2 - 4 SINGLE - P - A "weka.csv EuclideanDistance - R first-last"**

Cluster mode

- ☒ Use training set
- ☐ Supplied test set
- ☐ Percentage split
- ☐ Classes to clusters evaluation
- ☐ (Name) SpendingScore
- ☒ Store clusters for visualization

Ignore attributes

Start Stop

Result list (right-click for options)

- 06:38:09 - SimpleKMeans
- 06:38:50 - HierarchicalClusterer

Cluster output

ANALYZE CLUSTERING PROPERTIES (SUMMARY)

Cluster 0: 04,28000,45
Cluster 1: 02,69000,40

Missing values globally replaced with mean/mode

Final cluster centroids:

Attribute	Full Data	0	1
Customer	01	01	02
Income	46600	34000	62500
SpendingScore	54	45	57.5

Time taken to build model (Full training data) : 0.02 seconds

=== Model and evaluation on training set ===

Clustered Instances

0	3	(60%)
1	2	(40%)

Status: OK

Log

Weka Explorer

Preprocess Classify Cluster Associate

Cluster: **HierarchicalClusterer - N 2 - 4**

Cluster mode

- ☒ Use training set
- ☐ Supplied test set
- ☐ Percentage split
- ☐ Classes to clusters evaluation
- ☐ (Name) SpendingScore
- ☒ Store clusters for visualization

Ignore attributes

Start Stop

Result list (right-click for options)

- 06:38:09 - SimpleKMeans
- 06:38:50 - HierarchicalClusterer

Weka Cluster Visualizer: 06:38:09 - SimpleKMeans (customer_clusterer)

X Instance_number (Name)
Y Customer (Name)
Colour Cluster (Name)
Select Instance

Reset Clear Open Save Stop

Plot: customer_clusterer_clustered

Class colour

Status: OK

Log

Weka Explorer

Preprocess Classify Cluster Associate Select attributes Visualize

Open file...

Filter: **None**

Current relation: **None**

Relation: **None**

Instances: **None**

Attributes: **All**

Open

Look in: Downloads

Recent items

- agriculture
- book1
- roadBoard
- e-commerce
- employees
- hospital
- mobile
- Q1_Customer.arff
- supermarket
- telecom
- traffic

Invoke options dialog

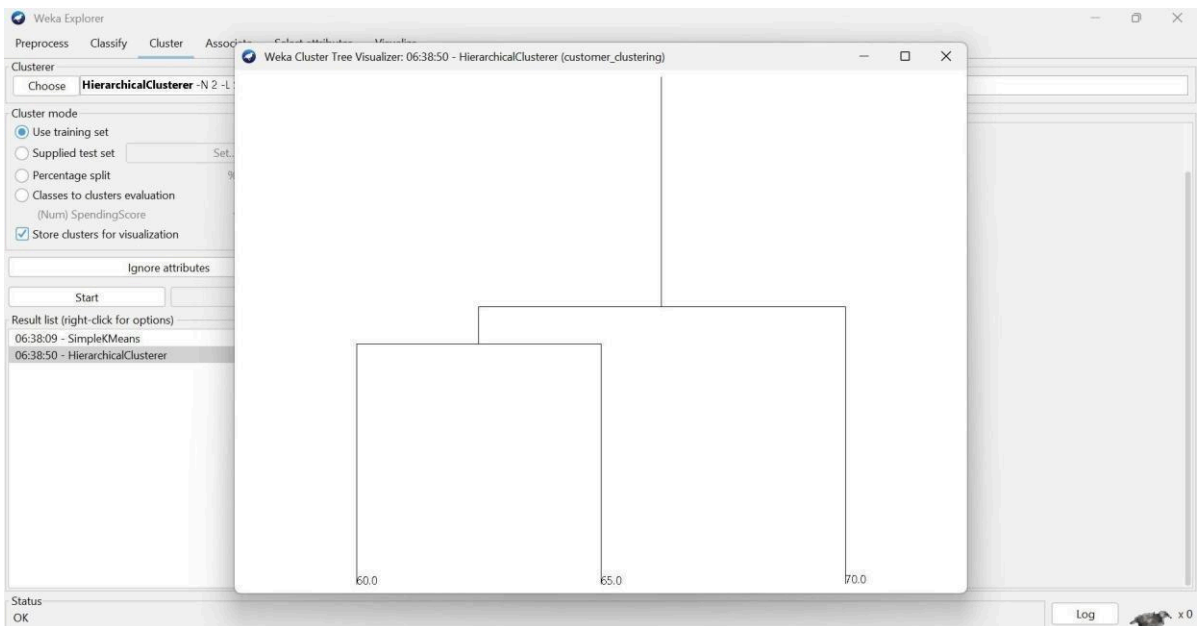
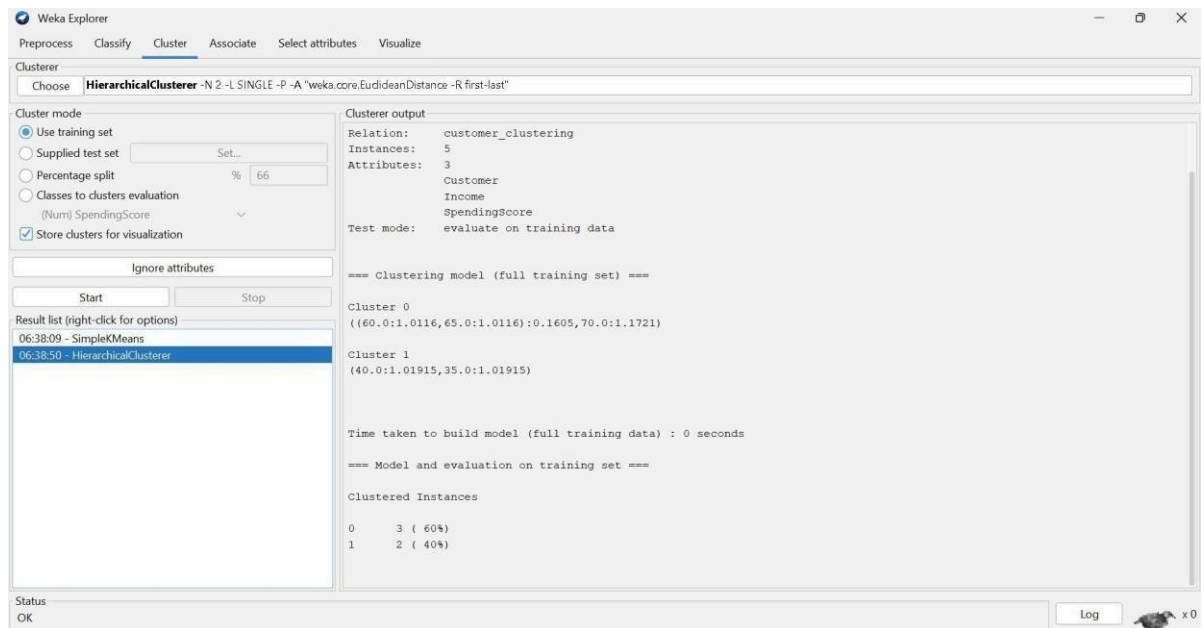
File Name: Q1_Customer.arff

File of type: Arff data files (*.arff)

Open Cancel

Status: Welcome to the Weka Explorer

Log



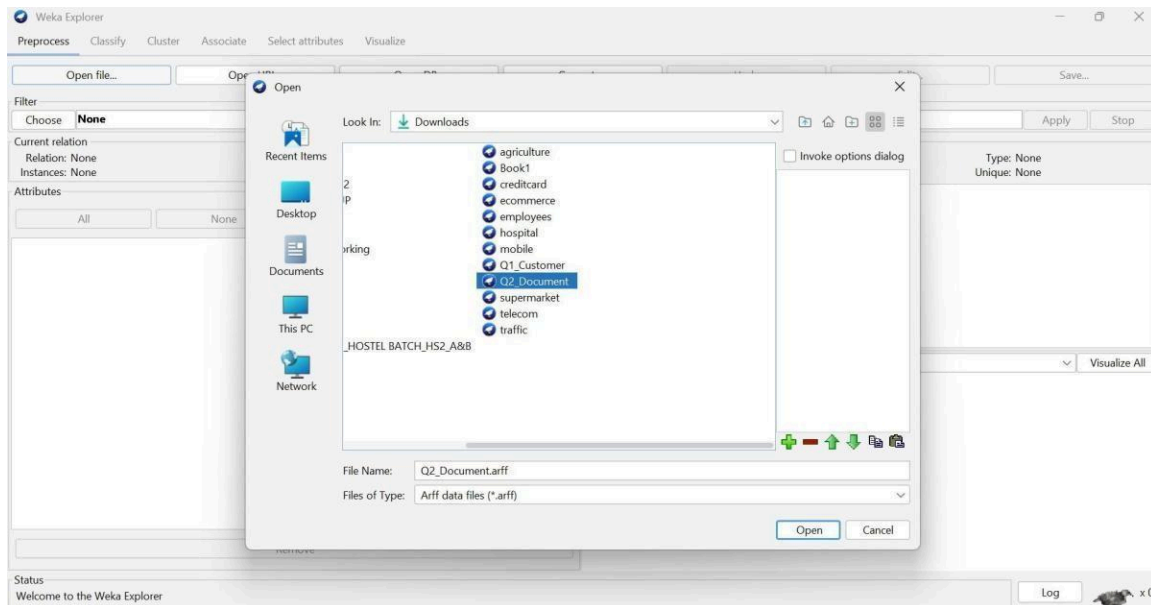
Q2.Dataset:DocumentData

DocumentWord1Word2Word3

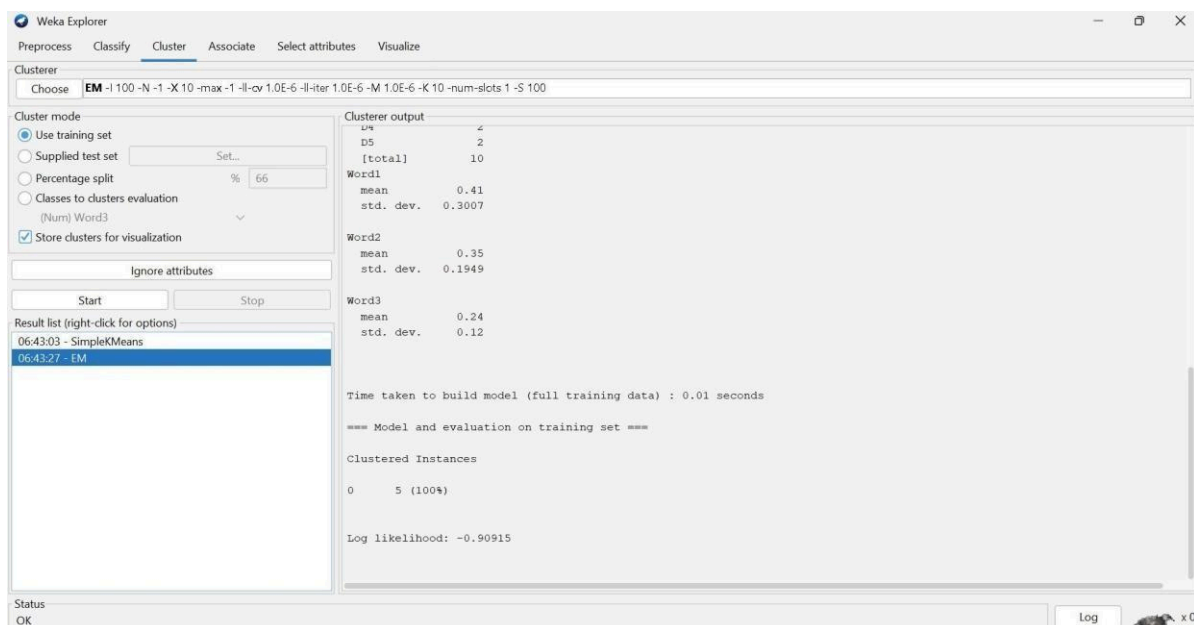
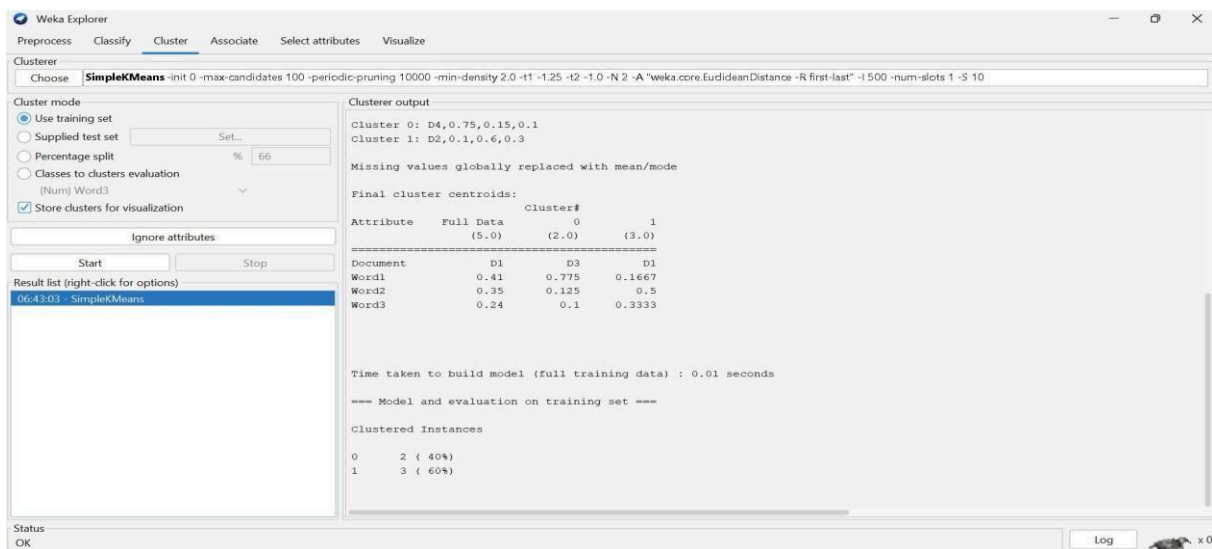
D1	0.2	0.5	0.3
D2	0.1	0.6	0.3
D3	0.8	0.1	0.1
D4	0.75	0.15	0.1
D5	0.2	0.4	0.4

Question:

Compare K-means and EM algorithms in handling overlapping clusters for the given dataset.
(BL4-1 Mark)



OUTPUT:



Weka Explorer
Preprocess Classify **Cluster** Associate Select attributes Visualize

Clusterer: Choose **SimpleKMeans** init 0 -max-candidates 100 -periodic-pruning 10000 -min-density 2.0 -t1 -1.25 -t2 -1.0 -N 2 -A "weka.core.EuclideanDistance -R first-last" -I 500 -num-slots 1 -S 10

Cluster mode:
☒ Use training set
☐ Supplied test set Set...
☐ Percentage split % 66
☐ Classes to clusters evaluation (Num) Word3
☒ Store clusters for visualization

Ignore attributes: []

Start Stop

Result list (right-click for options):
 06:43:03 - SimpleKMeans

Cluster output:

Cluster 0: D4,0.75,0.15,0.1
 Cluster 1: D2,0.1,0.6,0.3

Missing values globally replaced with mean/mode

Final cluster centroids:

Attribute	Full Data (5.0)	Cluster# 0 (2.0)	Cluster# 1 (3.0)
Document	D1	D3	D1
Word1	0.41	0.775	0.1667
Word2	0.35	0.125	0.5
Word3	0.24	0.1	0.3333

Time taken to build model (full training data) : 0.01 seconds

=== Model and evaluation on training set ===

Clustered Instances

Cluster	Count	Percentage
0	2	(40%)
1	3	(60%)

Status: OK Log

Weka Explorer
Preprocess Classify **Cluster** Associate Select attributes Visualize

Clusterer: Choose **EM** -I 100 -N -1 -K 10 -max -1 -R -or 1.0E-6 -R -or 1.0E-6 -M 1.0E-6 -K 10 -num-slots 1 -S 100

Cluster mode:
☒ Use training set
☐ Supplied test set Set...
☐ Percentage split % 66
☐ Classes to clusters evaluation (Num) Word3
☒ Store clusters for visualization

Ignore attributes: []

Start Stop

Result list (right-click for options):
 06:43:03 - SimpleKMeans
 06:43:27 - EM

Cluster output:

EM

Word	mean	std. dev.
Word1	0.41	0.3007
Word2	0.35	0.1949
Word3	0.24	0.12

Time taken to build model (full training data) : 0.01 seconds

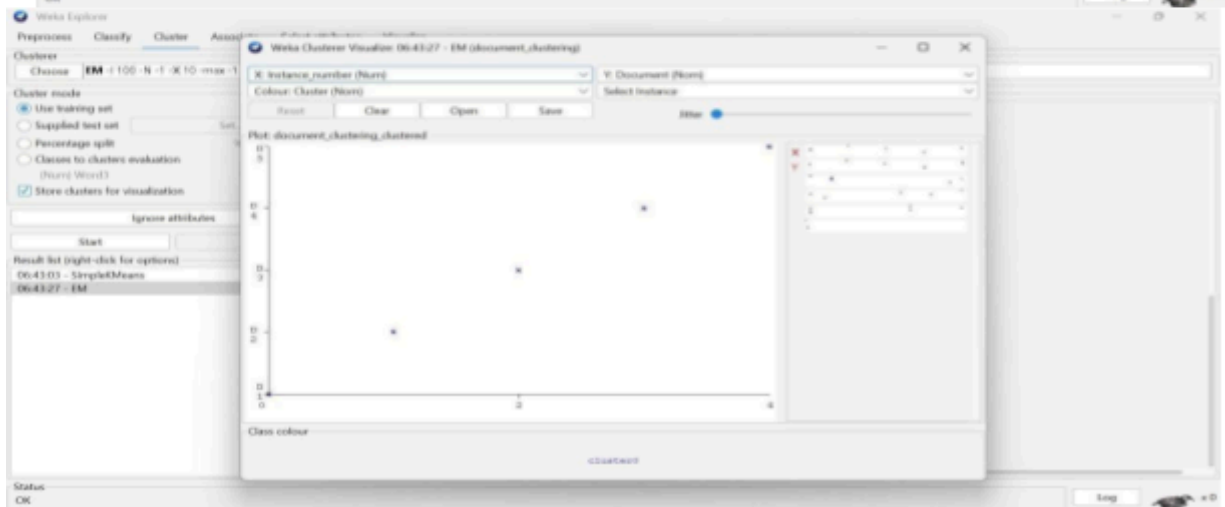
=== Model and evaluation on training set ===

Clustered Instances

Cluster	Count	Percentage
0	5	(100%)

Log likelihood: -0.90915

Status: OK Log



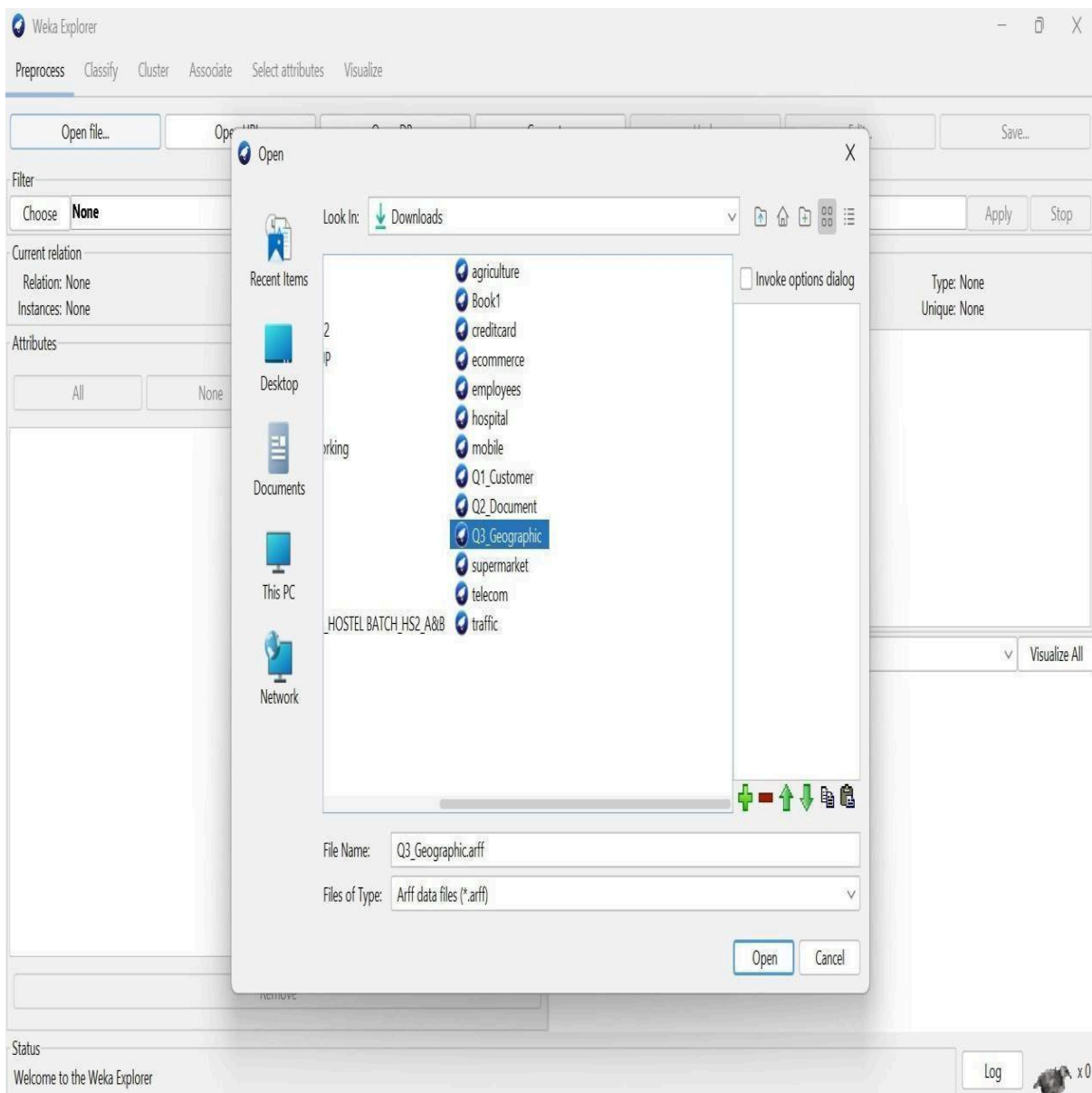
Q3.Dataset:Geographic Data

PointLatitudeLongitude

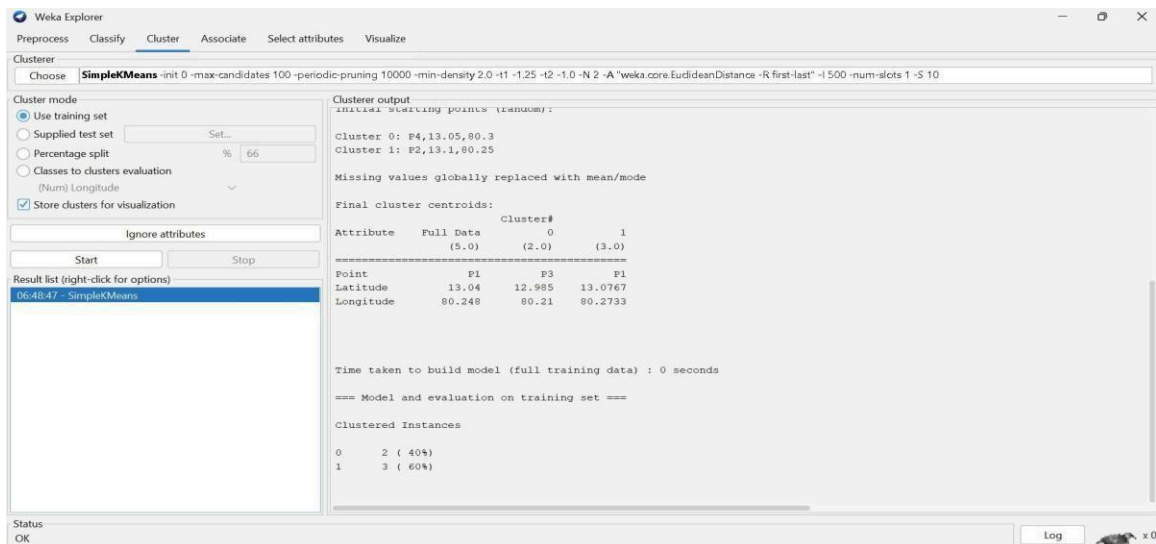
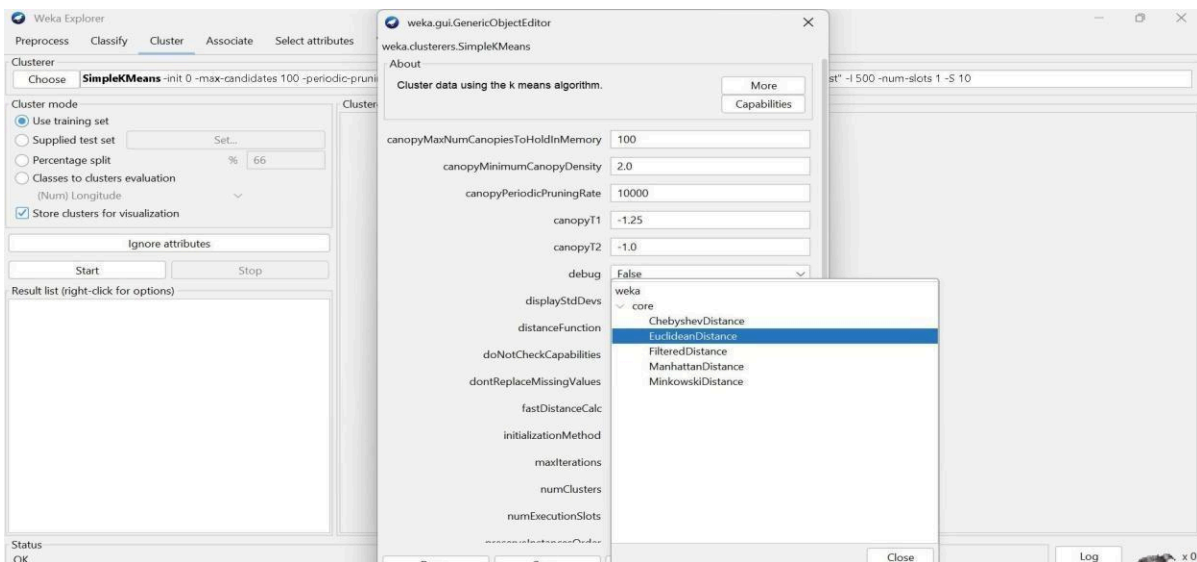
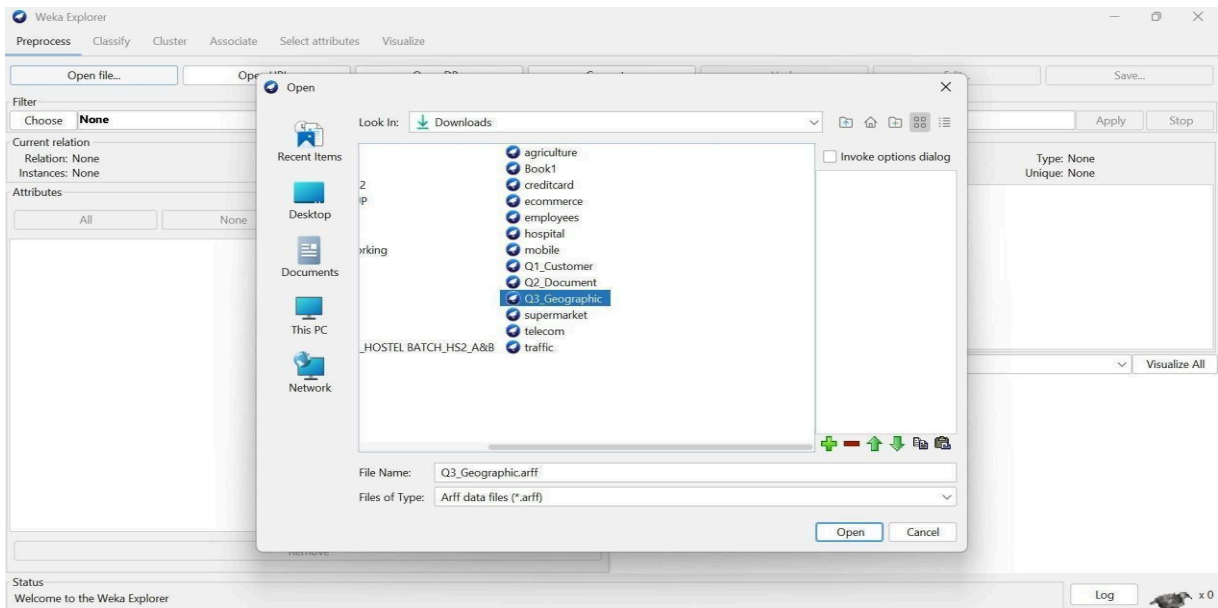
P1	13.08	80.27
P2	13.10	80.25
P3	12.97	80.22
P4	13.05	80.30
P5	13.00	80.20

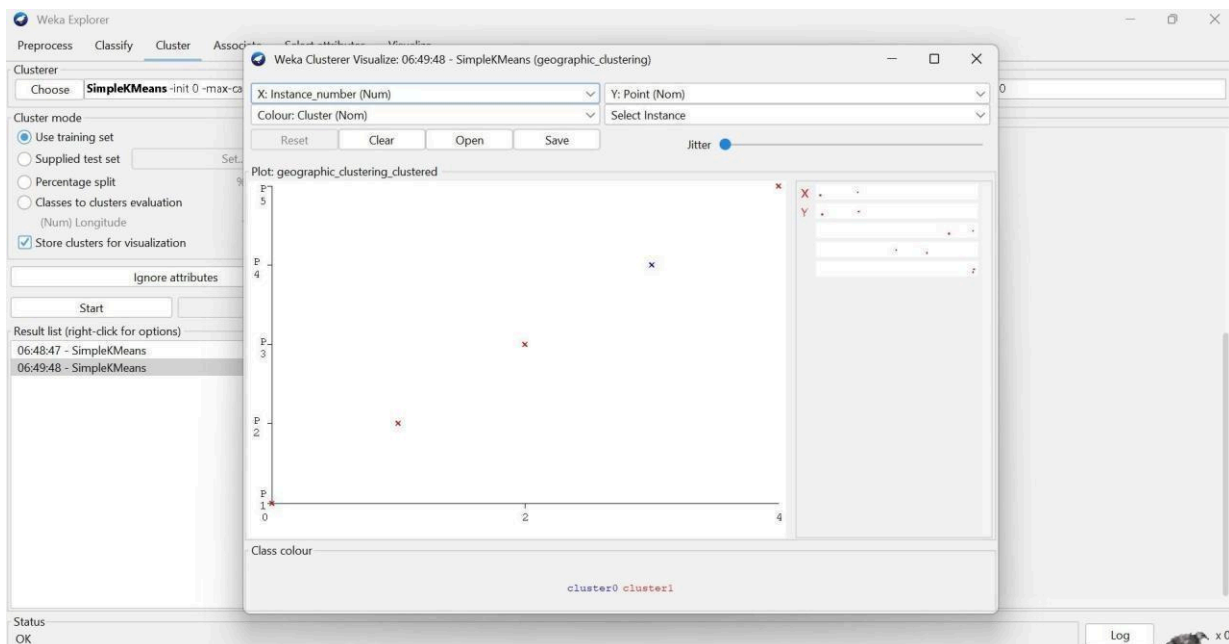
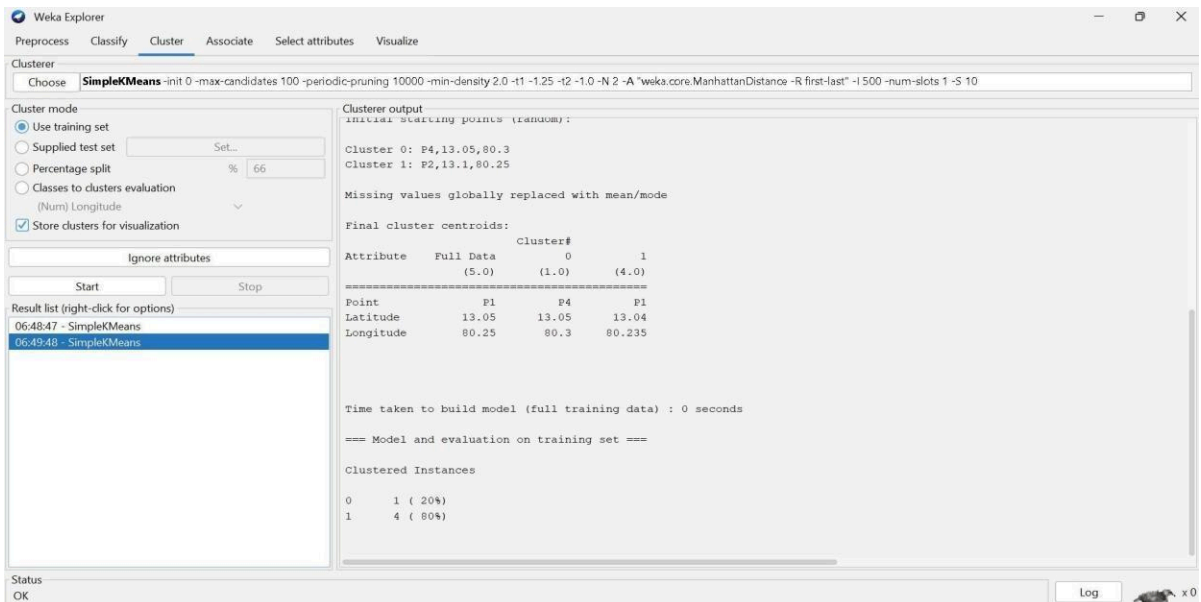
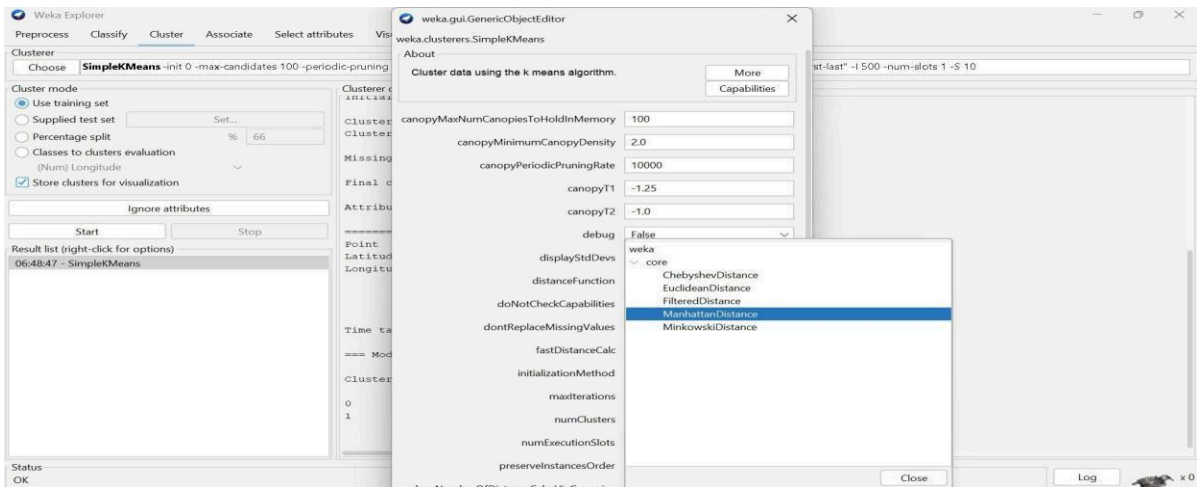
Question:

CompareEuclideanandManhattandistancemetricsandanalyzetheirimpactonclustering results.
(BL4 – 1 Mark)



OUTPUT:





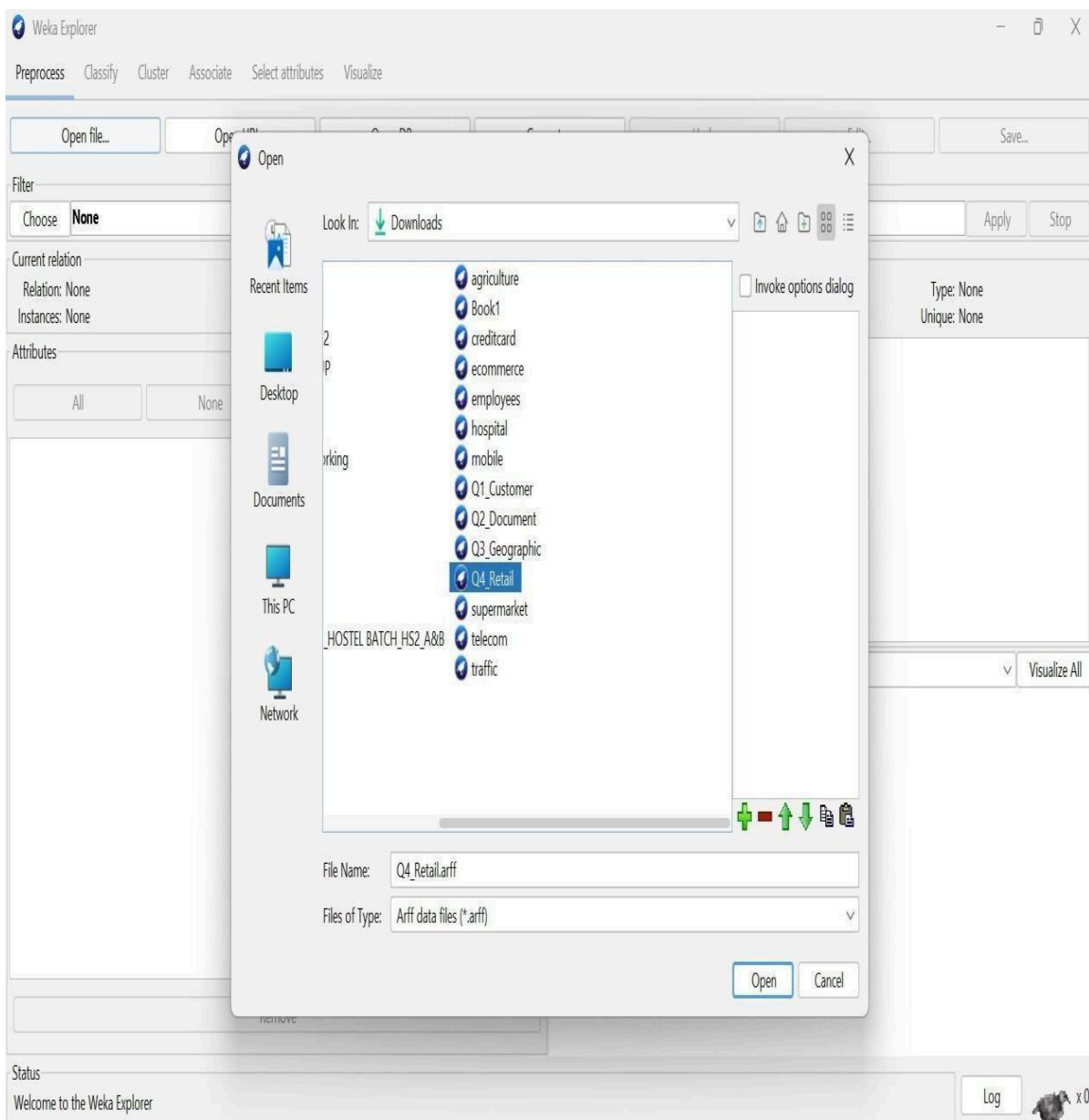
Q4.Dataset:RetailProducts

ProductSales(₹)Frequency

P1	50000	20
P2	30000	10
P3	52000	22
P4	25000	8
P5	32000	12

Question:

Compare agglomerative and divisive clustering approaches and explain their suitability for this dataset. (BL3 – 1 Mark)



OUTPUT:

The screenshot shows the Weka Explorer application window. The 'Cluster' tab is selected in the top menu. The 'Clusterer' dropdown is set to 'HierarchicalClusterer -N 2 -L SINGLE -P -A "weka.core.EuclideanDistance -R first-last"'. In the 'Cluster mode' section, 'Use training set' is selected, and 'Store clusters for visualization' is checked. The 'Ignore attributes' field is empty. The 'Start' button is visible. The 'Result list' on the left shows a single entry: '06:53:39 - HierarchicalClusterer'. The 'Clusterer output' pane on the right displays the following information:

Relation: retail_products
Instances: 5
Attributes: 3
Product
Sales
Frequency
Test mode: evaluate on training data

=== Clustering model (full training set) ===

Cluster 0
(20.0:1.01286,22.0:1.01286)

Cluster 1
((10.0:1.01286,12.0:1.01286):0.01412,8.0:1.02699)

Time taken to build model (full training data) : 0 seconds

=== Model and evaluation on training set ===

Clustered Instances

0	2 (40%)
1	3 (60%)

The status bar at the bottom shows 'Status OK' and a 'Log' button.

Weka Explorer

Preprocess Classify Cluster Associate Select attributes Visualize

Cluster

Choose **MakeDensityBasedClusterer** -M 1.0E-6 -W weka.clusterers.SimpleKMeans --init 0 -max-candidates 100 -periodic-pruning 10000 -min-density 2.0 -t1 -1.25 -t2 -1.0 -N 2 -A "weka.core.EuclideanDistance -R first-last" -I 500 -n

Cluster mode

☒ Use training set

☐ Supplied test set Set...

☐ Percentage split % 66

☐ Classes to clusters evaluation (Num) Frequency

☒ Store clusters for visualization

Ignore attributes

Start Stop

Result list (right-click for options)

06:53:39 - HierarchicalClusterer

06:54:50 - MakeDensityBasedClusterer

Cluster output

```
discreteEstimator.Counts = 1 2 1 2 2 (Total = 8)
Attribute: Sales
Normal Distribution. Mean = 29000 StdDev = 2943.9203
Attribute: Frequency
Normal Distribution. Mean = 10 StdDev = 1.633

Cluster: 1 Prior probability: 0.4286

Attribute: Product
Discrete Estimator. Counts = 2 1 2 1 1 (Total = 7)
Attribute: Sales
Normal Distribution. Mean = 51000 StdDev = 1000
Attribute: Frequency
Normal Distribution. Mean = 21 StdDev = 1

Time taken to build model (full training data) : 0 seconds

=== Model and evaluation on training set ===

Clustered Instances

0      3 ( 60%)
1      2 ( 40%)

Log likelihood: -12.6953
```

Status OK Log x0

Weka Explorer

Preprocess Classify Cluster Associate Select attributes Visualize

Cluster

Choose **MakeDensityBasedClusterer**

Cluster mode

☒ Use training set

☐ Supplied test set Set...

☐ Percentage split %

☐ Classes to clusters evaluation (Num) Frequency

☒ Store clusters for visualization

Ignore attributes

Start

Result list (right-click for options)

06:53:39 - HierarchicalClusterer

06:54:50 - MakeDensityBasedClusterer

Weka Clusterer Visualize: 06:54:50 - MakeDensityBasedClusterer (retail_products)

X: Instance_number (Num) Y: Product (Nom)

Colour: Cluster (Nom) Select Instance

Reset Clear Open Save Jitter

Plot: retail_products_clustered

Class colour

cluster0 cluster1

Status OK Log x0

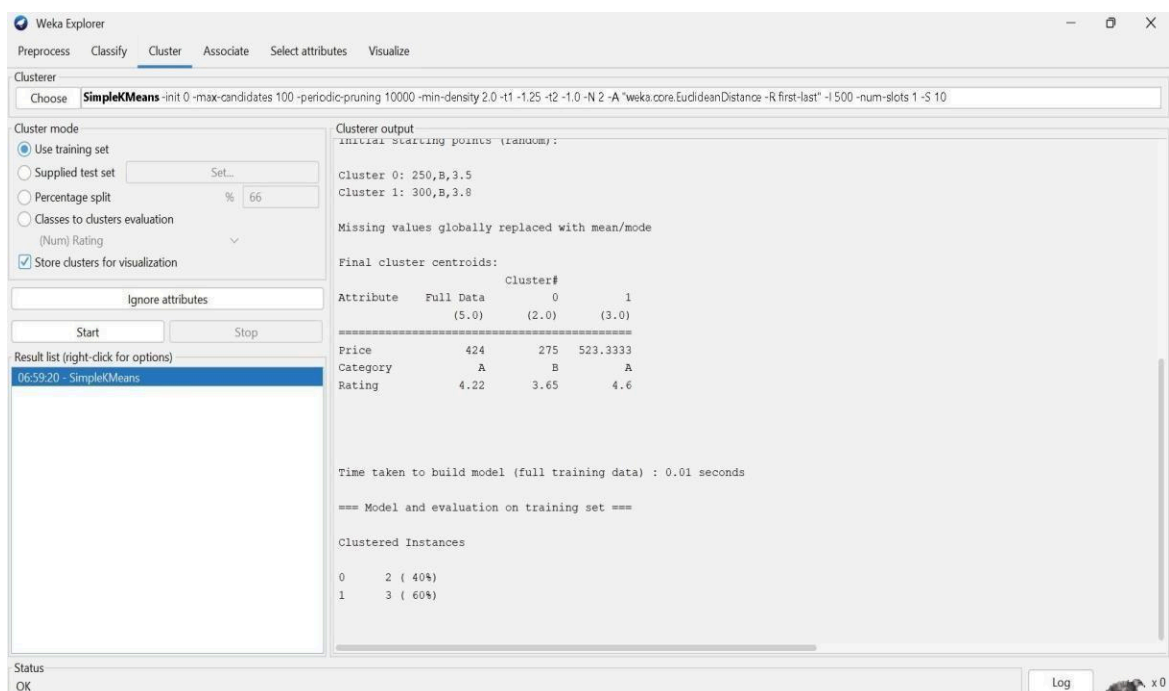
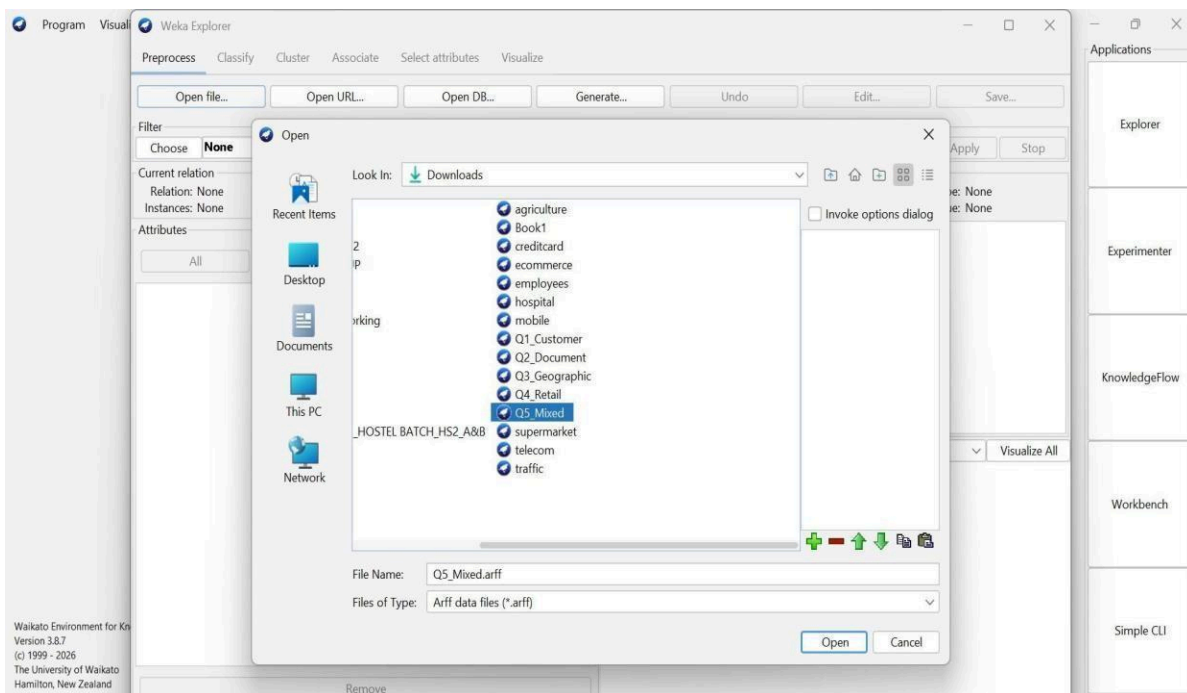
Q5.Dataset:MixedAttributes

ItemPrice(₹)CategoryRating

I1	500	A	4.5
I2	300	B	3.8
I3	550	A	4.7
I4	250	B	3.5
I5	520	A	4.6

Question:

Compare centroid-based and density-based clustering techniques and analyze their effectiveness on mixed data types. (BL4 – 1 Mark)



OUTPUT:

